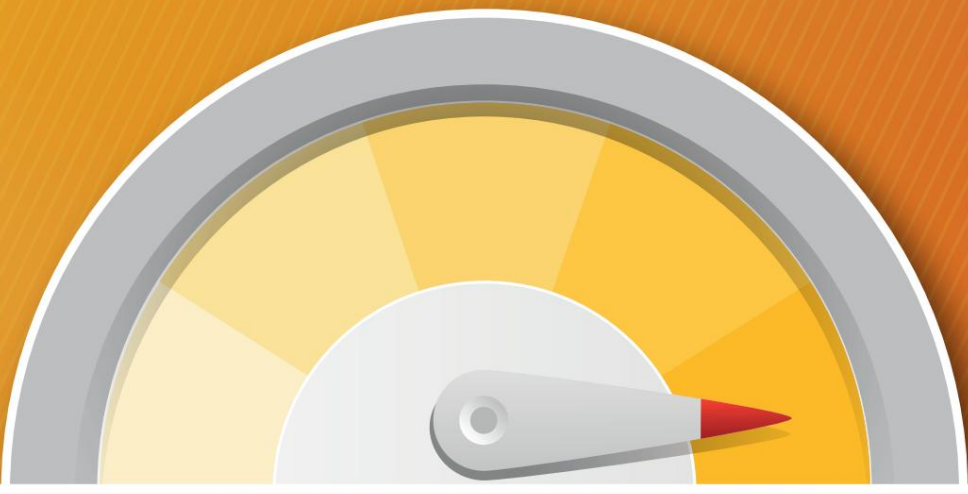




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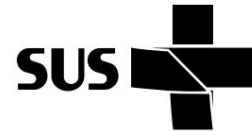


# PROTOCOL OF ADDRESSING **EXTREME HEAT**

1st EDITION



SAÚDE



# PROTOCOLO DE ENFRENTAMENTO AO **CALOR EXTREMO**

1st EDITION

Rio de Janeiro/RJ  
2024



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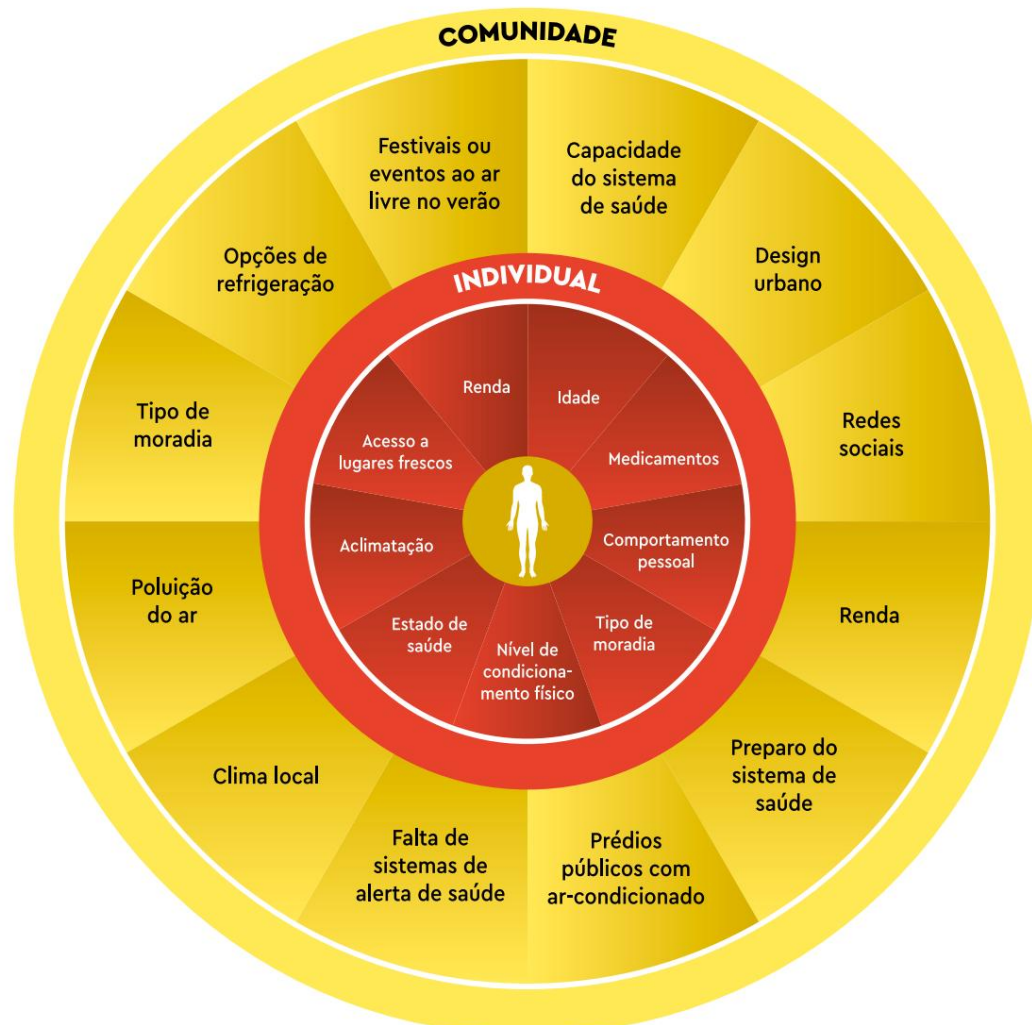
# Heat waves and their impacts on HUMAN HEALTH

Climate change represents one of the greatest challenges of the 21st century, affecting human health. Rising temperatures, extreme weather events, changes in precipitation patterns, and sea level rise have direct and indirect impacts on public health. Cities, with their high population density and economic activity, are especially vulnerable, requiring resilient and adaptive health systems.

Direct impacts include increased heat waves, which can cause dehydration, exacerbation of chronic diseases, heatstroke, and, in extreme cases, death. Indirectly, changes in precipitation and temperature patterns can alter the distribution of disease vectors, such as mosquitoes, increasing the incidence of vector-borne diseases like dengue. Food and water insecurity, exacerbated by climate change, can lead to malnutrition and related illnesses. Furthermore, there is damage to mental health and the way of life of territories, exacerbating existing vulnerabilities.

The effects of climate change are felt unevenly and tend to disproportionately affect lower-income areas. Vulnerable, historically marginalized populations living in the outskirts of large cities suffer more severely from the effects of extreme weather events. Therefore, considering social, racial, and environmental aspects is fundamental to developing actions that can mitigate inequalities among diverse population groups and promote more equitable health services and initiatives.

**Figure 1 — Factors influencing individual and community vulnerability to extreme heat events.**



During heat waves, most deaths are caused by the worsening of chronic conditions or an increase in infectious diseases. Clinical manifestations such as edema, rash, headache, irritability, lethargy, and weakness are common and vary according to the ability to adapt to extreme heat. Prolonged exposure to heat can cause dehydration, metabolic changes, burns, heat exhaustion, and eye and nasal irritation. In more extreme cases, it can lead to heatstroke, which is associated with high mortality.

Periods of intense heat affect the population, especially vulnerable groups such as children, the elderly, pregnant women, people with chronic illnesses, disabilities, mental disorders, outdoor workers, the homeless, migrants, refugees, and rural and indigenous communities. These groups are more susceptible to the impacts and can overburden health services due to the increased demand for care.

Identifying these higher-risk groups and situations allows for specific recommendations, protective measures, and intervention planning to mitigate health damage. With the increase and intensification of heat waves, it is crucial to develop coping strategies to reduce adverse health effects and ensure adequate care in these situations.

In this regard, the Municipal Health Department of Rio de Janeiro (SMS-Rio) presents the Protocol for Coping with Extreme Heat, with the main objective of guiding health actions during heat waves, aiming at protecting and mitigating health risks to the population.

This publication is part of the protocol of the City Hall of Rio de Janeiro (PCRJ) and is a pioneering and crucial initiative to protect the health of the population of Rio de Janeiro and ensure adequate care in these situations, in a context of increasingly frequent extreme weather events.

## CITY HEAT PROTOCOL RIO DE JANEIRO

The City Hall of Rio de Janeiro, through Decree Rio No. 54,740 of June 28, 2024 (Annex 1), established the classification of Heat Levels (HL) in risk scenarios related to extreme heat for the definition of action protocols within the municipality. The work, conducted by the Rio Operations and Resilience Center (COR), the Municipal Health Secretariat (SMS-Rio), and the Secretariat of Environment and Climate (SMAC-Rio), established metrics for the HL (Annex 2). These metrics take into account the heat index (HI), which results from the combination of temperature and humidity, as well as the exposure time. The COR is responsible for monitoring and disseminating the HL.

The NC (Normative Coordination) aims to define guidelines for action for various governmental bodies at different levels of government, public service concessionaires, and companies that provide services within the city of Rio de Janeiro, as well as for communication with citizens, as a parameter for triggering actions to address the climate risk of extreme heat. It is worth highlighting the pioneering role of the city of Rio de Janeiro, being the first in the country to implement an operational protocol for extreme heat based on health risks, integrating this range of bodies and the operation and functioning of the city.

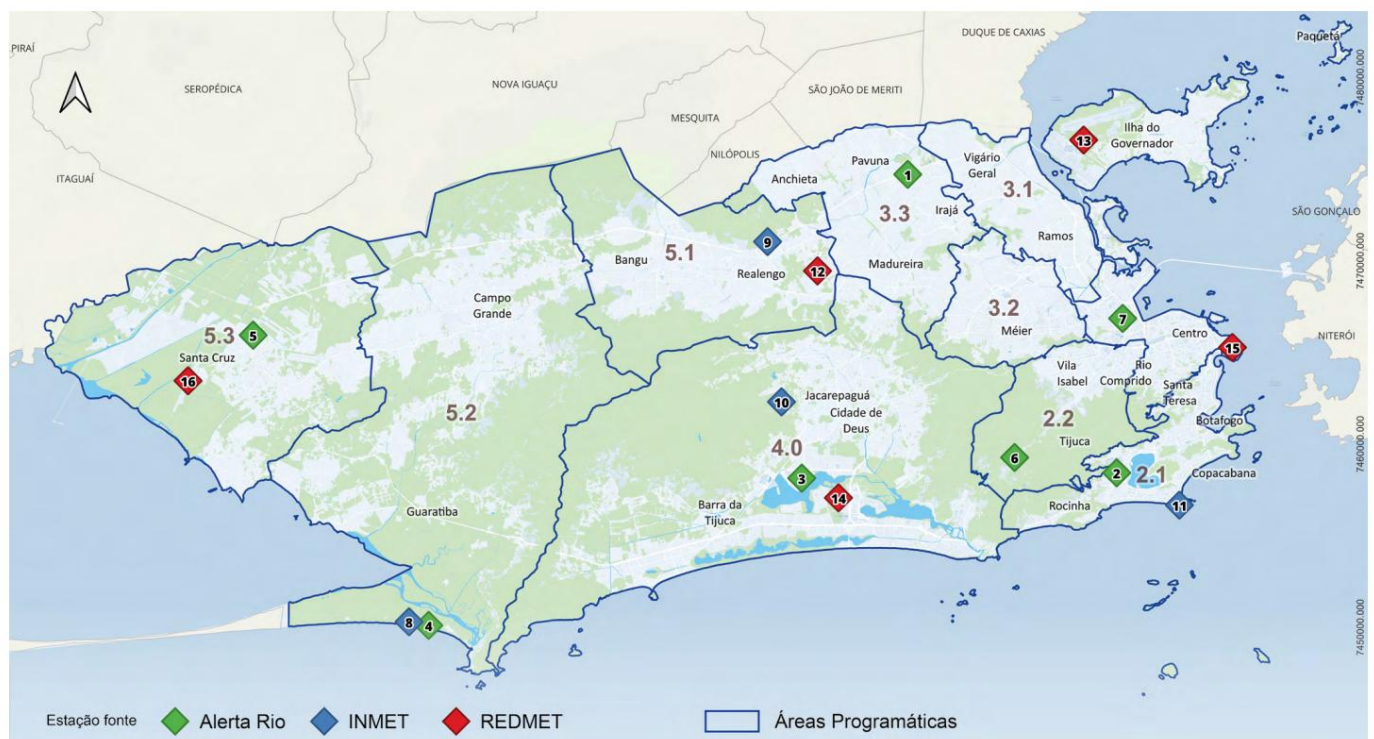
This decree also created, in the city of Rio de Janeiro, the Committee for the Development of Protocols for Coping with Extreme Heat (CDPECE), which will be responsible for defining coping protocols for risk scenarios related to extreme heat. This committee is composed of the COR, SMS-Rio and SMAC-Rio.

## CLIMATE AND HEALTH — EPIDEMIOLOGICAL INTELLIGENCE APPLIED TO EVENT PREPARATION AND RESPONSE EXTREME HEAT

To develop a method for establishing control thresholds for heat levels in light of health impacts, the Epidemiological Intelligence Center of the Superintendence of Health Surveillance of the Sub-Secretariat for Promotion, Primary Care and Health Surveillance (CIE/SVS/SUBPAV) carried out a series of analyses.

Initially, an extensive review of the scientific literature was conducted to identify the main impacts of extreme heat on human health. In parallel, data from meteorological stations of the municipal network Alerta Rio, the Meteorological Network of the Air Force Command (REDMET), and the National Institute of Meteorology (INMET) were used to construct time series of maximum, median, average, and minimum temperatures, humidity, and the IC (Index of Climate), applying statistical modeling to calculate historical quantiles of these conditions in the city of Rio de Janeiro.

**Figure 2 — Map showing the location of meteorological stations in the city of Rio de Janeiro.**



Source: CIE/AlertaRio/INMET/RedMet. Geographic Reference System.

Based on these quantiles, the effects of heat on human health were analyzed, considering care provided in the Emergency and Urgent Care Network (RUE), Primary Health Care (APS), and general mortality data from the Mortality Information System (SIM). Exploratory analyses identified a significant increase in health problems related to heat during extreme heat events in the city, or up to three days after such events.

Based on this analysis, and considering the risks of impacts on human health, it was possible to establish the heat levels to serve as the basis for the protocol of the City Hall of Rio de Janeiro.

Table 1 — Heat levels in the city of Rio de Janeiro

| LEVELS OF HEAT | CRITERION                                                                                                                                                                                                                                                                                              |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NC 1           | A level at which an IC (intravascular pressure) of less than or equal to 36°C (thirty-six degrees <i>Celsius</i> ) is recorded, or when it is greater than 36°C for a period of less than 4 (four) hours per day.                                                                                      |
| NC 2           | Level at which there is a recorded heart rate above 36°C for at least 4 (four) hours per day for up to 2 (two) consecutive days.                                                                                                                                                                       |
| NC 3           | Level in which there is a recorded IC between 36°C and up to 40°C for at least 4 (four) hours per day for at least 3 (three) consecutive days.                                                                                                                                                         |
| NC 4           | Level where there is a recorded heart rate between 40°C and up to 44°C for at least 4 (four) hours per day. In addition, there is a forecast or record of 2 (two) more days with a heart rate greater than 36°C for 4 (four) hours per day. The period must total at least 3 (three) consecutive days. |
| NC 5           | Level where there is a recorded heart rate above 44°C for at least 2 (two) hours per day. In addition, there is a forecast or record of 2 (two) more days with a heart rate greater than 36°C for 4 (four) hours per day. The period must total at least 3 (three) consecutive days.                   |

Sources: Rio Decree No. 54,740, of June 28, 2024, and Ordinance “N” No. 03/2024, of June 28, 2024.

It is worth highlighting that SMS-Rio, through the Health Surveillance CIE, continues to invest efforts in researching these findings to generate more evidence on this topic and maintains monitoring of epidemiological patterns, in a mosaic surveillance approach, based on the integration of information from a wide variety of information systems, aiming to support strategic decision-making for coordinated preparation and response to these events.

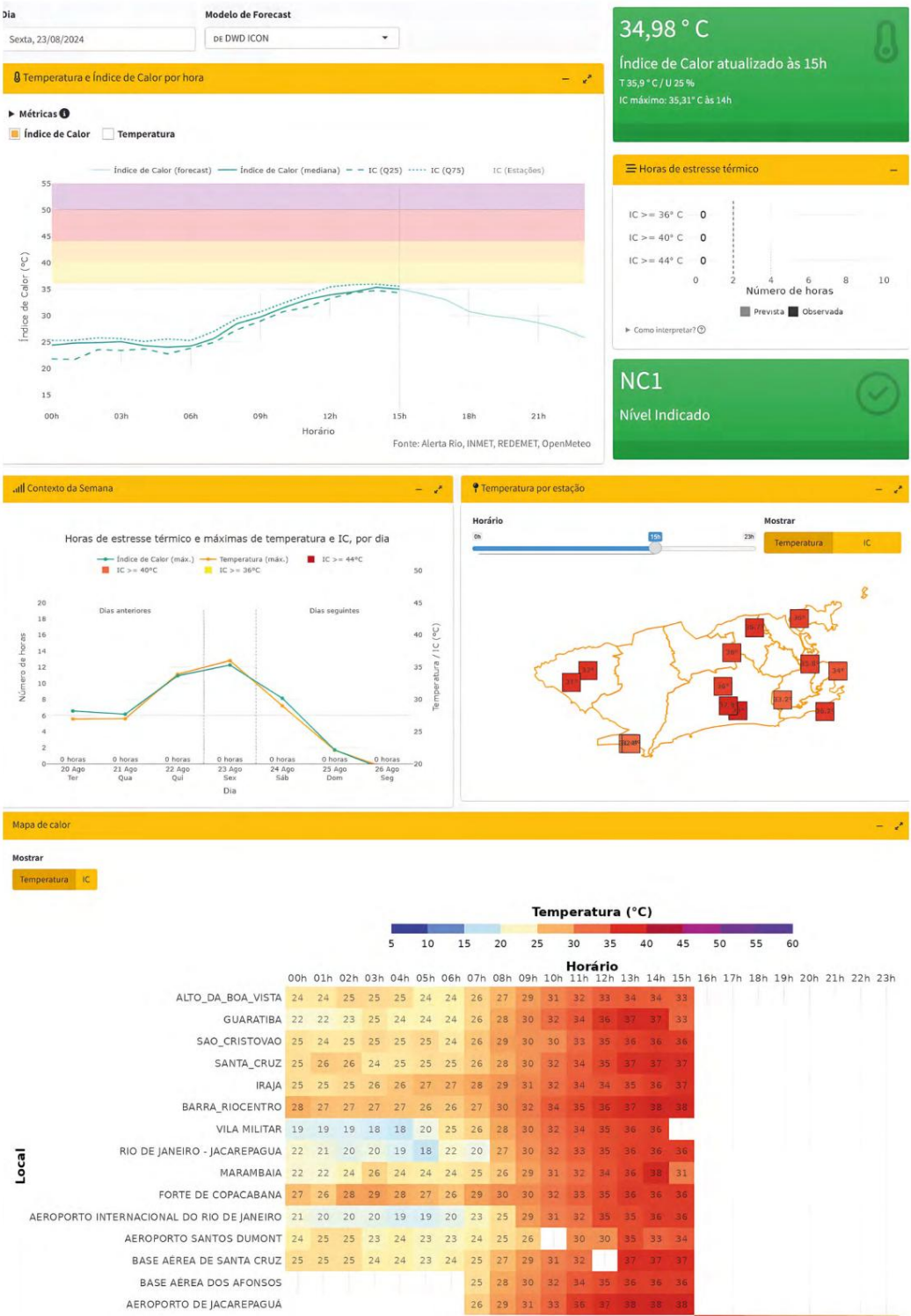
## Heat Panel — Monitoring and alerts based on epidemiological intelligence

Preparation for and response to extreme heat needs to be timely, and to enhance this capacity, the CIE (Center for Epidemiological Information) adopts a strategy of automatic retrieval of meteorological data from the Alerta Rio, INMET, and REDEMETS systems, as well as global meteorological models for numerical prediction (*forecast*) of temperature and humidity, in addition to data on care provided in the Emergency Care Network (RUE), Primary Health Care (APS), and hospital admission census. This large volume of data is entered into the CIE's *data lake*, processed, and updated daily in real time. Therefore, during an extreme heat event, it is possible to continuously measure the impact of care provided in order to prevent overloading the healthcare network and direct actions based on epidemiological intelligence.

In order to support decision-making, both in the context of public health and urban management and resilience, the CIE developed an analytical and interactive dashboard based on this data for use by the COR (Rio Operations Center), SMS-Rio (Rio de Janeiro Municipal Health Department), and strategic managers of the Rio de Janeiro City Hall. This dashboard integrates meteorological data with six predictive models from major international agencies, including the European Centre for Medium-Range Climate Predictions, the National Oceanic and Atmospheric Administration (NOAA), the German Weather Service (DWD), and Environment and Climate Change Canada. The dashboard automatically calculates the heat level based on the measured Heat Index (HI), exposure hours, and model forecasts, making it an essential tool for managing risks associated with extreme heat.

Finally, the integration of these meteorological forecast models , data obtained from ground-based sensor stations, and monitoring of the healthcare network will allow for the issuance of early warnings about heat waves to prepare health services and communicate risk to the population, as well as the adaptation measures foreseen in the city's protocol.

Figure 3 — Heat Panel



Sources: Alerta Rio, REDMET, INMET and OpenMeteo. Developed by CIE/SVS/SUBPAV/S.

## The Health Care Network of the Rio de Janeiro Municipal Health Department

The Health Care Network (RAS) is composed of health actions and services of different technological densities and profiles that, when integrated, seek to guarantee comprehensive care for the population.

Primary Health Care (PHC), as a structuring axis of the Health Care Network, has capillarity throughout the territory and plays a central role in ensuring the coordination of care and that the patient receives the health care they need. For this to occur, communication flows between PHC and the different points of the health care network must be established.

The city of Rio de Janeiro has 239 primary healthcare units (PHCUs), including family clinics (FCs) and municipal health centers (MSCs), which have qualified multidisciplinary teams to manage the main health problems of the population, in addition to carrying out prevention and health promotion actions. PHCUs have adequate physical infrastructure and supplies for the management and hydration of mild cases. The unit closest to a citizen's residence can be found on the "Where to Receive Care" platform website. ([prefeitura.rio/ondeser atendido](https://prefeitura.rio/ondeser atendido)).

The Emergency and Urgent Care Network (RUE) includes Urgent Care Units (UPAs), Regional Emergency Centers (CER), and emergency or urgent care hospitals that are equipped to receive patients requiring urgent or emergency interventions who are treated in Primary Health Care (APS) and referred through the Zero Vacancy platform, referrals to specialized care through the Regulation System (SISREG) and the State Regulation System (SER), and patients seeking care on their own initiative.

It is worth noting that, depending on the event and the impact of the heat waves, the initial rescue and pre-hospital care, as well as transportation to a hospital unit, may also be the responsibility of the Fire Department, according to the Matrix of Commitments and Responsibilities of municipal agencies.

The city of Rio also has a robust Health Surveillance network that integrates services and actions aimed at controlling determinants, risks, and damages to the population's health, also through the prevention of risks caused by hygiene and sanitation problems in activities, services, and products of interest to health. It is responsible for collecting, analyzing, and disseminating epidemiological data to assist in decision-making regarding the prevention and control of diseases, health problems, and outbreaks; surveillance of deaths; response to public health emergencies; planning and implementation of vaccination programs; sanitary inspection; control of zoonoses; and investigation of outbreaks caused by foodborne illnesses.

Regarding the Health Surveillance Network, actions are hierarchical and decentralized across the city's 10 planning areas, encompassing the entire Health Care Network, through operational, tactical, and strategic levels; centralized, regional, or local, comprising 10 Health Surveillance divisions, 28 Health Surveillance services, 10 Sentinel Surveillance units, 20 Rapid Response units, 28 hospital-based Epidemiological Surveillance centers, and 239 Primary Care units.

The table below lists the health units that make up the RAS (Health Care Network) under the management of the Municipality of Rio de Janeiro (MRJ).

Table 2 — Municipal health facilities providing care during heat waves

| COMPONENT                                                                                                         | SCOPE OF ACTION                                                                                                                                                    | NUMBER OF UNITS |
|-------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| <b>Care Unit<br/>Primary Health Care: Family<br/>Clinic (FC) and Center<br/>Municipal Health Department (CMS)</b> | Unit responsible for primary health care and coordination of all care.<br>Preferred entry point into the health and employment system with territorial allocation. | 239             |
| <b>Care Unit<br/>Primary Prison Unit (UAPP)</b>                                                                   | Primary Care Unit responsible for the care of the incarcerated population.                                                                                         | 28              |
| <b>Care Center<br/>Psychosocial (CAPS)</b>                                                                        | Monitoring of patients with severe mental disorders and/or addiction, requiring more intensive follow-up.                                                          | 38              |
| <b>Hospital</b>                                                                                                   | Emergency care for highly complex cases, including trauma and specialized care, and monitoring of patients who meet the criteria for hospital admission.           | 17              |
| <b>Motherhood and<br/>Birthing Center</b>                                                                         | Pay attention to childbirth.                                                                                                                                       | 14              |
| <b>Super Centro Carioca<br/>Health1</b>                                                                           | Conducting consultations and examinations, and specialized procedures.                                                                                             | 1               |
| <b>Super Centro Carioca<br/>Vaccination</b>                                                                       | Immunization.                                                                                                                                                      | 2               |
| <b>Polyclinic</b>                                                                                                 | Specialized outpatient care.                                                                                                                                       | 10              |
| <b>UPA (Emergency Care Unit)<br/>(Service) and CER (Regional<br/>Emergency Center)</b>                            | Urgent and emergency care.                                                                                                                                         | 15              |
| <b>Municipal Center<br/>Rehabilitation (CMR)</b>                                                                  | Diagnosis, assessment, guidance, early stimulation and specialized care in rehabilitation - provision, adaptation and maintenance of assistive technology.         | 2               |

Note: (1) The Super Centro Carioca de Saúde is composed of the Centro Carioca de Especialidades, the Centro Carioca de Diagnóstico e Tratamento por Imagem and the Centro Carioca do Olho. Source: CNES Platform — Competence June 2024, extracted on 07/30/2024.

# Disaster Risk Management and the IMPACTS OF HEAT LEVELS

Risk management is a continuous and permanent process that encompasses a set of actions aimed at preventing, reducing, and controlling, as much as possible, the risk factors present in the locality. Within the scope of Vigidesastres, risk management related to public health emergencies due to disasters mainly comprises action in the following phases: I – disaster preparedness; II – monitoring, alert, and communication; and III – response and rehabilitation (BRAZIL, 2022).

The Municipal Contingency Plan for Natural Disasters 2024-2026 was developed and published by SMS-Rio ([epirio.svs.rio.br/publicacoes](http://epirio.svs.rio.br/publicacoes)). This material is essential for risk management in health services, considering event typologies, disaster-related diseases, and new scientific evidence, ensuring an updated and effective approach. Heat waves and their possible consequences for human health fall under the typology of natural disasters, as these are described as events caused by natural processes or phenomena (hydrological, climatological, meteorological, and geological) that may imply human losses or other health impacts, damage to the environment and property, causing service interruptions and social and economic disruptions. Therefore, this protocol was constructed considering the same phases described in Vigidesastres (BRAZIL, 2022).

SMS-Rio organized its actions in each NC considering the disaster risk management described previously and the impacts on the health sector (Table 3).

Table 3 — Heat levels and their impacts on the health sector

| LEVELS OF HEAT | PHASES OF MANAGEMENT RISK OF DISASTERS      | RISK OF IMPACT ON THE HEALTH SECTOR                                                                                                                                                                                            |
|----------------|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NC 1           | Preparation                                 | There are no impacts. The focus is on ensuring that the entire healthcare network is aware of the protocols and is ready to implement actions at higher levels.                                                                |
| NC 2           | Monitoring, alerting and communication      | Increased risk for vulnerable groups. Intensified monitoring and issuance of alerts. Strengthened communication with the population and professionals.                                                                         |
| NC 3           | Monitoring, alert, communication and answer | Significant increase in risk and predicted impacts on human health. Intensification of communication campaigns and mitigation actions.                                                                                         |
| NC 4           | Response                                    | Occurrence of severe cases. Mobilization of the health network to operate protocols at maximum capacity. Evaluation of the suspension of external activities.                                                                  |
| NC 5           | Response and rehabilitation                 | Critical impacts on human health. Overburdened healthcare facilities. Adaptation and suspension of non-priority activities and redirection of efforts towards the protection and recovery of individuals affected by the heat. |

Source: Prepared by the author based on the Municipal Contingency Plan for Natural Disasters 2024-2026, SMS-Rio.

# CONTINGENCY PLAN FOR WAVES OF HEAT

The departments of SMS-Rio will act and adapt their actions in accordance with the established NC, prioritizing the protection of the population's health, considering actions related to assistance, adaptation of health services, surveillance, communication, and management.

## GUIDELINES AND OBJECTIVES

Table 4 — Guidelines and objectives of SMS-Rio according to heat levels

| LEVELS<br>HEAT | NC 1                                                                                                                                                             | NC 2                                                                                                                                                 | NC 3                                                                                                                                               | NC 4                                                                                                                                                                                                                                                                  | NC 5                                                                                                                                                                                     |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GOVERNANCE     | Prevention basic and awareness.                                                                                                                                  | Increased preventive measures and regular monitoring.                                                                                                | Reducing exposure and preparing for interventions more intense.                                                                                    | Implementation of adaptation and rapid response measures.                                                                                                                                                                                                             | Comprehensive emergency response and mitigation of extreme risks.                                                                                                                        |
| ASSISTANCE     | Maintain the routine activities of the units and encourage them in promotional actions, health, self-care in relation to exposure to the heat.                   | To educate the public about common clinical manifestations related to heat, with an emphasis on vulnerable groups, including those... warning signs. | Activate screening and heat-related condition management protocols at entry points and intensify monitoring of the most vulnerable groups.         | Activate hydration centers in primary health care units and intensify coordination between the health care network's services and other institutions to... the coordination of care.                                                                                  | Consider changes access resulting from high temperatures, with the aim of ensuring assistance and reducing the risk of heat stress for priority groups.                                  |
| INFRASTRUCTURE | Prepare and train healthcare teams on heat management and to ensure proper maintenance of equipment air conditioning and drinking fountains in the units health. | To ensure thermal comfort and hydration in healthcare facilities, with internal air conditioning.                                                    | To ensure thermal comfort and hydration in healthcare facilities, with internal air conditioning and means of communication with registered users. | To guarantee physical infrastructure and supplies for the centers. hydration. Adapt non-essential outdoor activities to an air-conditioned indoor environment. Implement measures for safety and well-being of the most vulnerable patients and professionals health. | Ensure that all entry points into the healthcare network are organized to address associated disorders. to the heat. Suspend non-priority external activities for professionals. health. |

| LEVELS     | HEAT | NC 1                                                                                                                                                                                                                            | NC 2                                                                                                                             | NC 3                                                                                                                                                                                         | NC 4                                                                                                                                                                                                                                                    | NC 5                                                                                                                                                                                                 |
|------------|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| HEALTH     |      | Maintain the heat alert system and routine surveillance actions.                                                                                                                                                                | Monitor heat indices, identify high-risk groups, and track heat-related illnesses. Maintain routine surveillance activities.     | To intensify disease mitigation, monitoring of the effects of heat, and the profiling of care provided. Issue alerts and communicate risks to the network.                                   | Establish a situation room at the Epidemiological Intelligence Center. Monitor morbidity and mortality patterns.<br><br>Adapt non-essential outdoor activities that involve direct exposure to heat and maintain teams of<br><br>Quick, ready response. | Maintain the guidelines from the previous level and structure the Emergency Operations Center.<br><br>Publish alerts and bulletins and reorganize surveillance services as needed.                   |
| HEALTH     |      | To guide services regarding prevention and self-care measures in relation to heat exposure, especially in nursing homes, therapeutic communities, schools, hospitals, events of sanitary interest, and collective environments. | To continue the actions from the previous level, aiming to mitigate the increased health risk.<br><br>because of the heat.       | Conduct sanitary inspections focusing on the health risks associated with heat exposure and ensure that establishments adopt preventive measures, especially for the most vulnerable groups. | Strengthen inspection efforts in healthcare settings and essential services, such as hospitals, nursing homes, and therapeutic communities.                                                                                                             | Maintain the guidelines from the previous level and expand inspection actions in areas of higher health risk.                                                                                        |
| COMMUNITY  |      | Munir a population of information. To build trust with the press and the public.                                                                                                                                                | Continue the actions from previous levels, work on internal communication, and monitor rumors to contain them.<br><br>fake news. | To continue the actions from the previous levels, expanding the reach and speed of message flow.                                                                                             | To continue the actions of the previous levels, disseminating information. to save lives.                                                                                                                                                               | To continue the actions from the previous levels, disseminating information to save lives. Evaluate and adjust the actions.                                                                          |
| MANAGEMENT |      | Maintain the responsiveness of the units and disseminate the protocol. of heat.                                                                                                                                                 | Ensure actions are taken according to the contingency plan and oversee the continuous monitoring of heat levels.                 | To oversee the effectiveness of the measures and update the contingency plan.                                                                                                                | Monitor the operational capacity of Network and preparedness for immediate action.                                                                                                                                                                      | Evaluate and consider the implementation of the Emergency Operations Center. Monitor response capacity and the need to expand beds or hydration centers according to the magnitude of the situation. |

# Contingency actions based on heat levels

## Heat level 1

Table 5 — Contingency actions by axis for heat level 1

| HEAT LEVEL 1                                        |                                                                                                                                                                                                                     |
|-----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AXLES                                               | ACTIONS                                                                                                                                                                                                             |
| ASSISTANCE<br>AND PREVENTION                        | To guide users on self-care measures, including avoiding sun exposure during peak hours and ensuring adequate fluid intake, as actions to promote health.                                                           |
|                                                     | Maintain available and up-to-date care protocols for heat-related illnesses and conditions.                                                                                                                         |
|                                                     | To promote and maintain the professional development of staff to manage heat-related illnesses and guide preventive measures.                                                                                       |
|                                                     | To promote and monitor educational actions for health promotion, such as the School Health Program and the Health RAP, addressing issues such as heat safety, hydration, and healthy and safe eating, among others. |
|                                                     | To identify the groups most vulnerable and susceptible to the effects of heat on health, and how such effects can vary in type, intensity, and magnitude.                                                           |
| INFRASTRUCTURE<br>AND ADAPTATION<br>OF THE SERVICES | Keep air conditioning units and water fountains in working order, positioned in strategic locations for easy access, with disposable cups available.                                                                |
|                                                     | Maintaining the supply of essential materials for patient care and laboratory tests.                                                                                                                                |
| SURVEILLANCE<br>IN HEALTH                           | Establish and maintain early warning systems for extreme weather events such as heat waves and disease outbreak forecasting, identifying the groups at highest risk.                                                |
|                                                     | Maintain monitoring and surveillance of diseases and health problems directly or indirectly related to heat.                                                                                                        |
|                                                     | To promote training, workshops, and simulations for professionals in the healthcare sector.                                                                                                                         |
|                                                     | Strengthen epidemiological surveillance of morbidity and mortality from heat-related illnesses.                                                                                                                     |
|                                                     | Intensify actions to prevent and control vectors, hosts, and reservoirs of diseases.                                                                                                                                |
|                                                     | To identify, validate, and map critical heat points in the territory.                                                                                                                                               |
| SURVEILLANCE<br>SANITARY                            | Maintain inspections and monitoring of Long-Term Care Facilities for the Elderly (ILPI) in order to verify ventilation, air conditioning, and air circulation.                                                      |
|                                                     | Maintain inspections and monitoring of Renal Replacement Therapy (RRT) services in order to verify ventilation, air conditioning, and air circulation.                                                              |
|                                                     | Maintain inspections in healthcare facilities to verify the availability of potable water, ventilation conditions, and air conditioning.                                                                            |

| HEAT LEVEL 1  |                                                                                                                                                                                                                                             |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AXLES         | ACTIONS                                                                                                                                                                                                                                     |
| COMMUNICATION | Disseminate information about the heat level monitoring actions carried out by the Rio de Janeiro City Hall and the measures planned through press <i>releases</i> and/or videos and/or audio recordings and/or materials for social media. |
|               | To store strategic information about the heat protocol, healthcare units, and network services.                                                                                                                                             |
|               | Make informational materials about heat safety permanently available on digital platforms (website, corporate TV, and others).                                                                                                              |
|               | Provide informational material to health units with guidelines for prevention and health promotion, including tips on hydration and healthy and safe eating, among others.                                                                  |
|               | Make routine posts on SMS-Rio's social media channels with guidelines for prevention and health promotion.                                                                                                                                  |
|               | Develop and publish content in collaboration with the COR (Center for Operations and Research) regarding heat protocols.                                                                                                                    |
|               | Create standard operating procedures (SOPs) for communication campaigns for NC 4 and 5.                                                                                                                                                     |
|               | Evaluate communication efforts and adapt the communication strategy in the contingency plan, if necessary.                                                                                                                                  |
|               | Maintain a list of spokespeople and a <i>mailing list</i> of specialized journalists.                                                                                                                                                       |
|               | Draft notices and announcements for a potential emergency.                                                                                                                                                                                  |
| MANAGEMENT    | Maintaining the responsiveness of health units through physical infrastructure, human resources, and equipment.                                                                                                                             |
|               | Disseminate and keep the Extreme Heat Response Protocol updated, and liaise with other sectors regarding the impacts on human health.                                                                                                       |

## Heat level 2

Table 6 — Contingency actions by axis for heat level 2

| HEAT LEVEL 2                                                 |                                                                                                                                                                                                                                                                                        |
|--------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AXLES                                                        | ACTIONS                                                                                                                                                                                                                                                                                |
| <b>ASSISTANCE<br/>AND PREVENTION</b>                         | Activate the screening and management protocol for heat-related conditions for the most vulnerable groups.                                                                                                                                                                             |
|                                                              | Intensify educational actions to promote health in health units (reception areas or waiting rooms), addressing sun exposure precautions, hydration, and healthy and safe eating.                                                                                                       |
|                                                              | Reinforce guidelines for preventing heat-related injuries, using social media, waiting room spaces, and even clinical consultations conducted for other reasons.                                                                                                                       |
|                                                              | To educate the public about common clinical manifestations related to heat, including self-management and identification of warning signs.                                                                                                                                             |
|                                                              | To monitor the groups that are most vulnerable and susceptible to the health effects of heat.                                                                                                                                                                                          |
| <b>INFRASTRUCTURE<br/>AND ADAPTATION<br/>OF THE SERVICES</b> | To monitor the current capacity of the Health Care Network (RAS) to address health risks and outcomes resulting from high temperatures.                                                                                                                                                |
|                                                              | Organize healthcare facilities to be places of thermal comfort and hydration, with air-conditioned interior spaces in all areas.                                                                                                                                                       |
|                                                              | Keep air conditioning units and water fountains in working order, positioned in strategic locations for easy access, with disposable cups available.                                                                                                                                   |
|                                                              | Maintaining the supply of essential medical supplies for patient care and laboratory tests.                                                                                                                                                                                            |
| <b>SURVEILLANCE<br/>IN HEALTH</b>                            | Monitor the mobilization stages daily and report them to the surveillance network.                                                                                                                                                                                                     |
|                                                              | To identify health risk factors and monitor the highest-risk groups.                                                                                                                                                                                                                   |
|                                                              | To establish communication between health units and other agencies.                                                                                                                                                                                                                    |
|                                                              | Issue a warning about potential health risks.                                                                                                                                                                                                                                          |
|                                                              | Intensify epidemiological surveillance of diseases of interest (sentinel diseases).                                                                                                                                                                                                    |
| <b>SURVEILLANCE<br/>SANITARY</b>                             | Maintain inspections and monitoring of Long-Term Care Facilities for the Elderly (ILPI) in order to verify ventilation, air conditioning, and air circulation, as appropriate, in addition to evaluating the planning for fluid management of elderly residents in these institutions. |
|                                                              | Maintain inspections and monitoring of Renal Replacement Therapy (RRT) services, verifying ventilation, air conditioning, and air circulation, as needed for each case, in order to ensure thermal comfort during dialysis sessions, avoiding clinical repercussions (hypotension).    |
|                                                              | Maintain inspections in healthcare facilities to verify the availability of potable water, ventilation conditions, and air conditioning.                                                                                                                                               |

| HEAT LEVEL 2  |                                                                                                                                                                                                                                    |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AXLES         | ACTIONS                                                                                                                                                                                                                            |
| COMMUNICATION | Disseminate the monitoring actions of the water levels carried out by the Rio de Janeiro City Hall and the measures planned through press <i>releases</i> and/or videos and/or audio recordings and/or materials for social media. |
|               | Make informational materials about heat safety permanently available on digital platforms (website, corporate TV, and others).                                                                                                     |
|               | To develop and publish content in collaboration with the COR (Operations Center of Rio de Janeiro) regarding heat protocols and specific health guidelines for warmer days.                                                        |
|               | Evaluate communication efforts and adapt the communication strategy in the contingency plan, if necessary.                                                                                                                         |
|               | Monitor SMS-Rio's channels to identify <i>fake news</i> , rumors, and hoaxes about heat levels.                                                                                                                                    |
|               | Develop and publish content on SMS-Rio's social media channels with specific health guidelines regarding heat.                                                                                                                     |
|               | Intensify media monitoring in search of potential trends.<br><i>(clipping)</i> .                                                                                                                                                   |
| MANAGEMENT    | Maintain the necessary resources and keep the action and contingency plan updated, prepared for heat-related prevention and care actions.                                                                                          |
|               | To oversee the continuous monitoring of heat levels and the effectiveness of the measures taken, adjusting strategies as needed.                                                                                                   |

## Heat level 3

Table 7 — Contingency actions by axis for heat level 3

| HEAT LEVEL 3                                                 |                                                                                                                                                                                                                                                                                        |
|--------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AXLES                                                        | ACTIONS                                                                                                                                                                                                                                                                                |
| <b>ASSISTANCE<br/>AND PREVENTION</b>                         | Activate the screening and management protocol for heat-related conditions for vulnerable groups.                                                                                                                                                                                      |
|                                                              | Intensify educational actions to promote health in health units (reception areas or waiting rooms), addressing sun exposure precautions, hydration, and healthy and safe eating, among other topics.                                                                                   |
|                                                              | Reinforce guidelines for preventing heat-related injuries, using social media, waiting room spaces, and even clinical consultations conducted for other reasons.                                                                                                                       |
|                                                              | To educate the public about common clinical manifestations related to heat, including self-management and identification of warning signs.                                                                                                                                             |
|                                                              | Intensify monitoring of the most vulnerable groups and those most susceptible to the health effects of heat.                                                                                                                                                                           |
|                                                              | To promote the relocation of patients, who are in health services, to an environment with greater thermal comfort (shaded environment, lower temperature and cooled floor).                                                                                                            |
|                                                              | Reinforce training with the teams at the entry points of hospital emergency rooms, UPAs (Emergency Care Units), and CERs (Emergency Care Centers) regarding the identification of disorders associated with heat exposure — providing guidance on the use of appropriate ICD codes.    |
| <b>INFRASTRUCTURE<br/>AND ADAPTATION<br/>OF THE SERVICES</b> | Prioritize activities with vulnerable and exposed groups in an air-conditioned indoor environment.                                                                                                                                                                                     |
|                                                              | Maintain the organization of health units to ensure they are places of thermal comfort and hydration, with air-conditioned interiors.                                                                                                                                                  |
|                                                              | Keep air conditioning units and water fountains in working order, positioned in strategic locations for easy access, with disposable cups available.                                                                                                                                   |
|                                                              | Maintaining the supply of essential medical supplies for patient care and laboratory tests.                                                                                                                                                                                            |
| <b>SURVEILLANCE<br/>IN HEALTH</b>                            | Monitor the mobilization stages daily and report them to the surveillance network.                                                                                                                                                                                                     |
|                                                              | To identify health risk factors and monitor the highest-risk groups.                                                                                                                                                                                                                   |
|                                                              | To establish communication between health units and other agencies.                                                                                                                                                                                                                    |
|                                                              | Issue a warning about potential health risks.                                                                                                                                                                                                                                          |
|                                                              | Intensify epidemiological surveillance of diseases of interest (sentinel diseases).                                                                                                                                                                                                    |
| <b>SURVEILLANCE<br/>SANITARY</b>                             | Maintain inspections and monitoring of Long-Term Care Facilities for the Elderly (ILPI) in order to verify ventilation, air conditioning, and air circulation, as appropriate, in addition to evaluating the planning for fluid management of elderly residents in these institutions. |

| HEAT LEVEL 3             |                                                                                                                                                                                                                                                                                          |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AXLES                    | ACTIONS                                                                                                                                                                                                                                                                                  |
| SURVEILLANCE<br>SANITARY | Increase inspections and monitoring of Renal Replacement Therapy (RRT) services, verifying ventilation, air conditioning, and air circulation, as appropriate for each case, in order to ensure thermal comfort during dialysis sessions, avoiding clinical repercussions (hypotension). |
|                          | Increase inspections in healthcare facilities to verify the availability of potable water, ventilation conditions, and air conditioning.                                                                                                                                                 |
|                          | To verify the conditions of potable water supply and the conditions of access, identification, and service at health care facilities during inspections of large-scale events.                                                                                                           |
| COMMUNICATION            | Disseminate information about the heat level monitoring actions carried out by the Rio de Janeiro City Hall and the measures planned through press releases and/or videos and/or audio recordings and/or materials for social media.                                                     |
|                          | Disseminate information about the change in risk level, along with recommendations for precautions, including any necessary adjustments to high-risk activities.                                                                                                                         |
|                          | Disseminate informational materials online about heat safety.                                                                                                                                                                                                                            |
|                          | To provide informational material and recommended messages for health units and services to share with users.                                                                                                                                                                            |
|                          | Develop press releases about precautions to take in the heat.                                                                                                                                                                                                                            |
|                          | Create and make available FAQs for the press.                                                                                                                                                                                                                                            |
|                          | To develop and publish content in collaboration with the COR (Center for Operations and Research) regarding heat protocols, changes in levels, and specific health guidelines for warmer days.                                                                                           |
|                          | Post specific health guidelines about heat on the SMS-Rio social media channels, including information for higher-risk groups, reinforcing the importance of following the recommendations.                                                                                              |
|                          | Evaluate communication efforts and adapt the communication strategy in the contingency plan, if necessary.                                                                                                                                                                               |
|                          | Monitor SMS-Rio's channels to clarify <i>fake news</i> , rumors, and misinformation about heat levels, prioritizing responses to inquiries on the subject via social media or the press, and creating content to reinforce the message when necessary.                                   |
|                          | Increase media monitoring to identify potential trends ( <i>clipping</i> ).                                                                                                                                                                                                              |
| MANAGEMENT               | Maintain the necessary resources for operationalizing the heat-related contingency and response plan.                                                                                                                                                                                    |
|                          | To supervise the effectiveness of the measures adopted, adjusting communication and assistance strategies as needed.                                                                                                                                                                     |

## Heat level 4

Table 8 — Contingency actions by axis for heat level 4

| HEAT LEVEL 4                                        |                                                                                                                                                                                                                                                                                 |
|-----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AXLES                                               | ACTIONS                                                                                                                                                                                                                                                                         |
| ASSISTANCE<br>AND PREVENTION                        | Activate hydration centers for the population in primary care units.                                                                                                                                                                                                            |
|                                                     | Activate the screening and management protocol for heat-related conditions for the general population.                                                                                                                                                                          |
|                                                     | Strengthen coordination between the RAS (Regionalized Health System) units and other institutions, such as emergency services, schools, daycare centers, and community organizations, in order to ensure a timely and coordinated response to the effects of high temperatures. |
|                                                     | Monitor healthcare pressure and requests for Zero Bed availability based on ICD codes for heat-related conditions, and adjust bed allocation.                                                                                                                                   |
|                                                     | Prioritize scheduling appointments for vulnerable groups and those susceptible to the effects of heat, during times of less intense temperatures.                                                                                                                               |
|                                                     | Utilize telehealth resources when available.                                                                                                                                                                                                                                    |
|                                                     | To assess the possibility of home visits to populations at higher risk of illness or worsening of diseases due to high temperatures, especially children under 2 years of age and elderly people with impaired basic activities of daily living.                                |
|                                                     | To keep patients in healthcare facilities in a more thermally comfortable environment (shaded environment, lower temperature, and cooled floor).                                                                                                                                |
|                                                     | To guide patients on resources in the area for relief during heat waves (publicly accessible air-conditioned locations, as well as swimming pools or strategic water distribution centers in the city).                                                                         |
| INFRASTRUCTURE<br>AND ADAPTATION<br>OF THE SERVICES | Strengthen the organization of teams at the entry points of emergency hospitals, UPAs (Emergency Care Units), and CERs (Emergency Care Centers) to address disorders associated with heat exposure — providing guidance on the use of appropriate ICD codes.                    |
|                                                     | To ensure the physical structure and adequate supplies for activating hydration points in primary care units.                                                                                                                                                                   |
|                                                     | Prioritize activities in air-conditioned indoor environments, suspending, whenever possible, activities in unshaded outdoor areas.                                                                                                                                              |
|                                                     | Keep air conditioning units and water fountains in working order, with disposable cups readily available, positioned in strategic locations for easy access by people.                                                                                                          |
|                                                     | Maintaining the supply of essential medical supplies for patient care and laboratory tests.                                                                                                                                                                                     |

| HEAT LEVEL 4                                        |                                                                                                                                                                                                                                                                                           |
|-----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AXLES                                               | ACTIONS                                                                                                                                                                                                                                                                                   |
| INFRASTRUCTURE<br>AND ADAPTATION<br>OF THE SERVICES | Take measures to ensure the safety and well-being of healthcare professionals: maintain adequate hydration, wear light and comfortable clothing, and take protective measures against solar radiation, such as using sunscreen, hats, and sunglasses.                                     |
|                                                     | Activate a hydration and refreshment break for professionals who are engaged in outdoor activities, such as the Street Clinic teams, at least every two hours, or whenever necessary, in a shaded area, for hydration and application of sunscreen.                                       |
| SURVEILLANCE<br>IN HEALTH                           | To carry out a supplementary assessment of the damage and impacts on health and to evaluate changes in the morbidity and mortality pattern.                                                                                                                                               |
|                                                     | Prepare the climate and health bulletin with the results of the monitoring indicators from SMS-Rio (Rio de Janeiro Municipal Health Department) related to health care services.                                                                                                          |
|                                                     | Prioritize indoor activities in air-conditioned environments for surveillance professionals and adopt hydration breaks for priority health surveillance activities that cannot be adapted, postponed, or suspended.                                                                       |
|                                                     | To detect outbreaks early and implement actions for the immediate control of heat-related communicable and non-communicable diseases, such as: waterborne and foodborne illnesses, vector-borne diseases, cardiovascular diseases, and accidents involving venomous animals.              |
|                                                     | Maintain the network of Rapid Response teams on standby for the need for action and epidemiological investigation, considering the NC (National Curriculum).                                                                                                                              |
|                                                     | To make available specific antivenoms for accidents involving venomous animals, given the variability of this type of accident depending on temperature conditions.                                                                                                                       |
| SURVEILLANCE<br>SANITARY                            | Maintain inspections and monitoring of Long-Term Care Facilities for the Elderly (ILPI) in order to verify ventilation, air conditioning, and air circulation, as appropriate, in addition to evaluating the planning for fluid management of elderly residents in these institutions.    |
|                                                     | Intensify inspections and monitoring of Renal Replacement Therapy (RRT) services, verifying ventilation, air conditioning, and air circulation, as appropriate for each case, in order to ensure thermal comfort during dialysis sessions, avoiding clinical repercussions (hypotension). |
|                                                     | Intensify inspections in healthcare facilities to verify the availability of potable water, ventilation conditions, and air conditioning.                                                                                                                                                 |
|                                                     | Increase the number of teams inspecting large-scale events, verify the availability of drinking water, and assess access, identification, and service conditions at healthcare facilities.                                                                                                |

| HEAT LEVEL 4  |                                                                                                                                                                                                                                                        |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AXLES         | ACTIONS                                                                                                                                                                                                                                                |
| COMMUNICATION | Disseminate information about the heat level monitoring actions carried out by the Rio de Janeiro City Hall and the measures planned through press <i>releases</i> and/or videos and/or audio recordings and/or materials for social media.            |
|               | Disseminate information about the change in risk level, along with recommendations for precautions, including any necessary adjustments to high-risk activities.                                                                                       |
|               | Increase the dissemination of online informational materials about heat safety.                                                                                                                                                                        |
|               | To provide informational material and recommended messages for health units and services to share with users.                                                                                                                                          |
|               | Disseminate specific materials to vulnerable groups, taking comorbidities into account.                                                                                                                                                                |
|               | Prepare and distribute <i>press releases</i> to the media <i>mailing list</i> regarding precautions against the heat. Evaluate the possibility of a press conference.                                                                                  |
|               | Create and make available FAQs for the press.                                                                                                                                                                                                          |
|               | Create video material demonstrating how heat levels work.                                                                                                                                                                                              |
|               | To develop and publish content in collaboration with the COR (Center for Operations and Research) regarding heat protocols, level changes, and specific health guidelines for warmer days.                                                             |
|               | Post specific health guidelines about heat on the SMS-Rio social media channels, including information for higher-risk groups, reinforcing the importance of following the recommendations.                                                            |
|               | Monitor SMS-Rio's channels to clarify <i>fake news</i> , rumors, and misinformation about heat levels, prioritizing responses to inquiries on the subject via social media or the press, and creating content to reinforce the message when necessary. |
|               | Establish partnerships with public service institutions/companies, such as Me-trô Rio and CCR Barcas, to disseminate informational messages.                                                                                                           |
|               | Develop materials for street furniture with specific health guidelines regarding heat.                                                                                                                                                                 |
|               | Evaluate communication efforts and adapt the communication strategy in the contingency plan, if necessary.                                                                                                                                             |
|               | Establish a partnership with Civil Defense to send out SMS messages with instructions.                                                                                                                                                                 |
|               | Intensify media monitoring in search of potential trends.<br>( <i>clipping</i> ).                                                                                                                                                                      |
|               | Strengthen internal communication with units for early notification of potential spontaneous agendas and specific cases.                                                                                                                               |
|               | Seeking individuals (patients and professionals) to suggest story ideas about the effects of heat, drawing media attention to the issue.                                                                                                               |

| HEAT LEVEL 4 |                                                                                                                                                                                                                                                                                                           |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AXLES        | ACTIONS                                                                                                                                                                                                                                                                                                   |
| MANAGEMENT   | Implement and coordinate immediate actions to address heat-related health issues, ensuring that health facilities are prepared to serve the vulnerable population.                                                                                                                                        |
|              | To ensure collaboration between different technical areas and sectors involved, such as health, communication, infrastructure, assistance, and surveillance.                                                                                                                                              |
|              | Consider reducing the period of outdoor activities for health workers who work under direct sun exposure and solar radiation (e.g., home visits by Community Health Agents and Health Surveillance Agents) without the possibility of body cooling, suspending them during periods of highest heat index. |
|              | Healthcare workers who are part of priority groups for heat protection should perform their duties in the unit, to allow for the possibility of body cooling.                                                                                                                                             |
|              | Consider adapting the activities of the Academia Carioca Program to a location with the possibility of body cooling and lower physical intensity.                                                                                                                                                         |
|              | Consider implementing a health situation room.                                                                                                                                                                                                                                                            |

Source: Author's own elaboration, 2024.

## Heat Level 5

Table 9 — Contingency actions by axis for heat level 5

| HEAT LEVEL 5                 |                                                                                                                                                                                    |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AXLES                        | ACTIONS                                                                                                                                                                            |
| ASSISTANCE<br>AND PREVENTION | Maintain primary care units as hydration centers for the population.                                                                                                               |
|                              | Activate the screening and management protocol for heat-related conditions for the general population.                                                                             |
|                              | Maintain capacity to increase and meet spontaneous demand.<br>Prioritize scheduling for higher-risk groups during times of lower heat index.                                       |
|                              | Evaluate scheduling of planned appointments and/or consider telehealth services if available to expand capacity for spontaneous demands and avoid displacement of priority groups. |
|                              | Strengthen ties with the RAS (Regional Health Administration) in order to ensure a timely, coordinated, and longitudinal response to the effects of high temperatures.             |

| HEAT LEVEL 5                                        |                                                                                                                                                                                                                                                             |
|-----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AXLES                                               | ACTIONS                                                                                                                                                                                                                                                     |
| ASSISTANCE<br>AND PREVENTION                        | To guarantee timely and coordinated care for the entire population, consistent with the screening and management protocol for heat-related conditions.                                                                                                      |
|                                                     | To keep patients in healthcare facilities in a more thermally comfortable environment (shaded, cooler, and air-conditioned).                                                                                                                                |
|                                                     | Monitor the regulation and availability of hospital beds, expanding management and administration capabilities as needed.                                                                                                                                   |
|                                                     | Increase patient awareness and information about resources available in the area for relief during heat waves (publicly accessible air-conditioned facilities, as well as swimming pools or strategic water distribution centers in the city).              |
|                                                     | To ensure the organization of the entry points of emergency hospitals, UPAs (Emergency Care Units), and CERs (Emergency Care Centers) to absorb the demand for disorders associated with heat exposure.                                                     |
| INFRASTRUCTURE<br>AND ADAPTATION<br>OF THE SERVICES | Prioritize activities in air-conditioned indoor environments, suspending outdoor activities whenever possible.                                                                                                                                              |
|                                                     | Keep air conditioning units and water fountains in working order, positioned in strategic locations for easy access. Keep disposable cups readily available near the water fountains.                                                                       |
|                                                     | Maintaining the supply of essential medical supplies for patient care and laboratory tests.                                                                                                                                                                 |
|                                                     | Strengthen measures for the safety and well-being of healthcare professionals: maintain adequate hydration, wear light and comfortable clothing, and adopt protective measures against solar radiation, such as the use of sunscreen, hats, and sunglasses. |
|                                                     | Ensure that professionals on the Street Clinic teams take breaks every two hours, or whenever necessary, in a shaded area, for hydration and application of sunscreen.                                                                                      |
|                                                     | Monitor healthcare pressure in primary care, urgent and emergency care, and hospital beds, and adjust supply as needed.                                                                                                                                     |

| HEAT LEVEL 5              |                                                                                                                                                                                                                                                                                          |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AXLES                     | ACTIONS                                                                                                                                                                                                                                                                                  |
| SURVEILLANCE<br>IN HEALTH | To provide the Emergency Operations Center (EOC) with immediate and timely information about the situation, to support decision-making.                                                                                                                                                  |
|                           | Monitor emergency and urgent care network data more frequently in light of the heat index — regarding alerts.                                                                                                                                                                            |
|                           | Activate Rapid Response teams when an outbreak occurs or when the operational capacity of healthcare units increases, taking into account patient flow.                                                                                                                                  |
|                           | Review and adapt the preparedness and response plan, protocols, and procedures.                                                                                                                                                                                                          |
|                           | After NC 5, evaluate performance in preparation and response — lessons learned.                                                                                                                                                                                                          |
|                           | Reorganize the Health Surveillance services as needed.                                                                                                                                                                                                                                   |
|                           | Prepare and disseminate the climate and health bulletin after NC 5.                                                                                                                                                                                                                      |
| SURVEILLANCE<br>SANITARY  | Maintain inspections and monitoring of Long-Term Care Facilities for the Elderly (ILPI) in order to verify ventilation, air conditioning, and air circulation, as appropriate, in addition to evaluating the planning for fluid management of elderly residents in these institutions.   |
|                           | Increase inspections and monitoring of Renal Replacement Therapy (RRT) services, verifying ventilation, air conditioning, and air circulation, as appropriate for each case, in order to ensure thermal comfort during dialysis sessions, avoiding clinical repercussions (hypotension). |
|                           | Increase inspections in healthcare facilities to verify the availability of potable water, ventilation conditions, and air conditioning.                                                                                                                                                 |
|                           | Increase the number of teams inspecting large-scale events, verify the availability of drinking water, and assess access, identification, and service conditions at healthcare facilities.                                                                                               |

| HEAT LEVEL 5  |                                                                                                                                                                                                                                                        |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AXLES         | ACTIONS                                                                                                                                                                                                                                                |
| COMMUNICATION | Disseminate information about the change in risk level, along with recommendations for precautions, including any necessary adjustments to high-risk activities.                                                                                       |
|               | Increase the dissemination of online informational materials about heat safety.                                                                                                                                                                        |
|               | To provide informational material and recommended messages for health units and services to share with users.                                                                                                                                          |
|               | Disseminate specific materials to vulnerable groups, taking comorbidities into account.                                                                                                                                                                |
|               | Call a press conference to warn about health risks, signs of heat stress, and precautions for preventing health problems.                                                                                                                              |
|               | Create and make available FAQs for the press.                                                                                                                                                                                                          |
|               | Create video material demonstrating how heat levels work.                                                                                                                                                                                              |
|               | To develop and publish content in collaboration with the COR (Center for Operations and Research) regarding heat protocols, level changes, and specific health guidelines for warmer days.                                                             |
|               | Post specific health guidelines about heat on the SMS-Rio social media channels, including information for higher-risk groups, reinforcing the importance of following the recommendations.                                                            |
|               | Monitor SMS-Rio's channels to clarify <i>fake news</i> , rumors, and misinformation about heat levels, prioritizing responses to inquiries on the subject via social media or the press, and creating content to reinforce the message when necessary. |
|               | Establish partnerships with public service institutions/companies, such as Me-trô Rio and CCR Barcas, to disseminate informational messages.                                                                                                           |
|               | Develop materials for street furniture with specific health guidelines regarding heat.                                                                                                                                                                 |
|               | Establish a partnership with Civil Defense to send out SMS messages with instructions.                                                                                                                                                                 |
|               | Evaluate the communication efforts and results, and adapt the communication strategy in the contingency plan if necessary.                                                                                                                             |
|               | To work together with the City Hall's advertising department on possible advertising campaigns about the heat protocol.                                                                                                                                |

| HEAT LEVEL 5  |                                                                                                                                                                                                                                              |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AXLES         | ACTIONS                                                                                                                                                                                                                                      |
| COMMUNICATION | Strengthen internal communication with units for early notification of potential spontaneous agendas and specific cases.                                                                                                                     |
|               | Seeking individuals (patients and professionals) to suggest story ideas about the effects of heat, drawing media attention to the issue.                                                                                                     |
|               | Increase media monitoring to identify potential trends ( <i>clipping</i> ).                                                                                                                                                                  |
| MANAGEMENT    | To monitor the response actions and operational capacity of the services.                                                                                                                                                                    |
|               | Evaluate performance in preparedness and response, identifying lessons learned and adjusting preparedness plans, protocols, and procedures as needed.                                                                                        |
|               | Assess the need for and consider implementing an Emergency Operations Center (EOC) in light of the indicators presented.                                                                                                                     |
|               | Establish telehealth appointments scheduled during peak temperature hours to avoid unnecessary travel for groups at higher risk from heat-related issues.                                                                                    |
|               | Suspend the activities of the Academia Carioca Program and <i>on-site</i> group activities to avoid displacement of the target audience.                                                                                                     |
|               | Suspend non-priority outdoor activities for healthcare workers who work under direct sun exposure and solar radiation (e.g., home visits by Community Health Agents and Health Surveillance Agents) without the possibility of body cooling. |

# MAIN CONDITIONS AND RISKS OF EXTREME HEAT FOR HEALTH — GUIDELINES AND CLINICAL MANAGEMENT

## Preventive measures for the general population

Preventive measures are crucial to minimize the risks associated with heat and protect health. Among the basic precautions recommended are:

**Hydration:** Increase your intake of drinking water or natural fruit juices without added sugar, and avoid alcoholic, caffeinated, or sugary drinks, which can cause dehydration.

Water can be ingested in various ways: *in its natural state* and through foods rich in water. Most fruits contain between 80% and 90% liquid. In situations of extreme heat, it can be beneficial to ingest oral rehydration solutions or sports drinks with electrolytes to replace not only water but also salts lost through sweat. People with heart failure, high blood pressure, or kidney disease, and children should not take them without professional advice.

Consume a large volume of fluids, greater than what you lose through urine or sweat, without waiting until you are thirsty. The color of your urine is a good indicator: you should increase your fluid intake when your urine becomes darker.

**Diet:** Opt for light meals that are less fatty, with less salt and minimal seasoning, in other words, foods that are easy to digest (avoid ready-to-eat processed foods).

Prefer to eat smaller portions more frequently (small, light, refreshing meals with a high water content). Avoid consuming diuretic drinks and thermogenic foods and beverages (coffee, mate tea, green tea, cinnamon, ginger, peppers). Pay attention to food preparation and storage. Observe the hygiene conditions of the place and be careful when eating out. Intense heat accelerates the food degradation process and increases the risk of contamination; therefore, it is important to observe the storage location and temperature. Avoid foods that have been exposed to room temperature for more than two hours. Wash your hands thoroughly before starting food preparation and during the process whenever necessary. Properly sanitize raw foods (such as fruits, vegetables, and legumes) by washing them with running water and then using a chlorinated water solution (one tablespoon of sodium hypochlorite per liter of water). Thaw food in the refrigerator or in a microwave oven when the food will be cooked immediately. Refrigerate cooked and perishable foods.

**Appropriate clothing:** Wear light-colored, lightweight, and loose-fitting clothing that allows sweat to evaporate, helping to maintain body temperature.

**Sun protection:** Wear wide-brimmed hats, sunglasses with UV protection, and sunscreen with a sun protection factor (SPF) of at least 30, reapplying as needed.

**Cool environments:** Stay in cool, well-ventilated areas, using fans or air conditioning if available.

**Acclimation:** Gradually increasing exposure to heat to allow the body to adjust, reducing the risk of heat-related illnesses.

## Care for high-risk groups

The risk of heat-related illnesses results from a combination of individual susceptibility, exposure to endogenous and exogenous heat, and sociocultural factors that affect the ability to adapt.

Patients over 65 years of age, young children, infants, pregnant women, people with pre-existing medical conditions (such as obesity, cardiovascular disease, diabetes, respiratory diseases, kidney failure, and mental disorders), people with disabilities (PWD), outdoor workers, and athletes are at higher risk. In addition, people living in low-income households face elevated risks.

Table 10 below summarizes the main groups vulnerable to heat waves, along with their particularities. Table 11 lists medications that affect the body's response to heat.

Table 10 — Specific characteristics of risk groups in relation to the effects of heat.

| GROUPS OF GREATER RISK       | SPECIAL FEATURES                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Children                     | Infants, especially newborns and premature babies, have a limited capacity to dissipate heat due to the immaturity of their sweating mechanisms and their high body surface area to volume ratio. Older children, despite having a greater capacity for heat dissipation, still need guidance from caregivers to maintain hydration and adopt preventive measures against heat. It is crucial to ensure adequate hydration, the use of light clothing, and to avoid direct sun exposure. For breastfed children, advise the nursing mother to offer the breast more often, and for older children, who may not feel or show thirst, offer water, natural fruit juices, and fruits more frequently. Warning signs include: excessive thirst, reduced urination (fewer diaper changes in infants) and skin turgor, irritability, lethargy, dry mouth, and fatigue. |
| Elderly (> 65 years old)     | Older adults have a reduced capacity to dissipate heat due to decreased sweating and cutaneous blood flow, aggravated by reduced blood volume and increased baseline inflammatory state. These factors hinder thermoregulation and increase the risk of hyperthermia. This is compounded by the high prevalence of chronic conditions, often associated with multimorbidity, the use of medications that can interfere with thermoregulation, and a tendency towards lower fluid intake. Special attention should be given to frail older adults, who have a higher risk of clinical deterioration, therefore requiring differentiated attention from family and health services. Older adults may not feel thirsty. Offer water, natural fruit juices, and fruits more often, preferably in small portions and volumes, and more frequently.                    |
| People with chronic diseases | Chronic diseases, such as obesity, hypertension, diabetes <i>mellitus</i> , cardiovascular and respiratory diseases, significantly increase the risk of complications during heat stress events. This may result from the interference of the disease or the medications used to treat it (Table 11) in the response to heat, as well as from greater vulnerability to the effects of heat stress, especially dehydration. The particularities of some diseases are highlighted in Table 12.                                                                                                                                                                                                                                                                                                                                                                     |

| GROUPS OF GREATER RISK                                                  | SPECIAL FEATURES                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>People in homelessness</b></p>                                    | <p>For the homeless population, clinical vulnerability is exacerbated by social vulnerability. These individuals are more exposed to extreme heat due to reduced access to air-conditioned environments and sun protection. Furthermore, they face a high burden of morbidity from various socially determined clinical conditions and have less access to potable and chilled water, as well as adequate food.</p>                                                                                                                                                                                                                                                     |
| <p>People who perform professional activities</p> <p><b>outdoor</b></p> | <p>For outdoor workers, constant exposure to heat and sun significantly increases the risk of heat stress and accidents. The use of uniforms and PPE can intensify the sensation of heat, requiring special consideration in their selection to ensure ventilation and comfort. The cognitive consequences of heat stress increase the risk of accidents, and there is a greater risk of burns from contact with hot surfaces. Essential preventive measures include reinforcing hydration, taking frequent breaks, monitoring temperature and humidity, adapting functions for cooled environments, and adjusting work schedules to periods of lower temperatures.</p> |
| <p><b>Athletes who engage in vigorous exercise no heat</b></p>          | <p>For individuals who remain physically active during intense heat waves, it is recommended to modify exercise routines to minimize health risks. It is crucial to encourage regular hydration, suggest exercising during cooler times of day (early morning or evening), use air-conditioned facilities, reduce exercise intensity, and opt for lower-impact activities. It is also important to educate about the risks and provide clear guidance, as well as monitor and support with hydration stations and rest areas.</p>                                                                                                                                       |
| <p><b>Individuals with mobility restrictions and bedridden</b></p>      | <p>For bedridden individuals or those with reduced mobility, physical limitations hinder the search for cooler environments and access to hydration. Furthermore, this population frequently presents with chronic conditions, especially neurological ones, associated with reduced thermoregulation capacity. During heat waves, it is important that homes are adapted to ensure access to more thermally comfortable locations and hydration. Public spaces for heat relief should also be accessible to people with disabilities.</p>                                                                                                                              |
| <p><b>Pregnant women and women who are breastfeeding</b></p>            | <p>Pregnant women face an increased risk of premature birth, low birth weight, and stillbirth during heat waves. Dehydration and extreme heat impact the physiological changes of pregnancy, reducing uterine blood flow and fetal oxygenation, in addition to exacerbating emotional stress. During pregnancy, women already have higher body temperature and lower blood pressure, conditions aggravated by intense heat. Social inequities in pregnancy indicators are exacerbated, with a greater impact on low-income and Black women.</p>                                                                                                                         |

**Table 11 — List of medications and other substances that affect the body's response to heat.**

| <b>Medications and substances that alter the body's response to heat.</b>                                                                                                                                                 |                                                                                                                                                                                                       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Alcohol</b>                                                                                                                                                                                                            | It can reduce alertness and affect judgment and heat perception; it exacerbates dehydration and affects vasodilation and cardiac contractility.                                                       |
| <b>Amphetamines</b>                                                                                                                                                                                                       | They can increase metabolic heat production.                                                                                                                                                          |
| <b>Anticholinergics</b>                                                                                                                                                                                                   | They can reduce sweat production.                                                                                                                                                                     |
| <b>Antidepressants tricyclics</b>                                                                                                                                                                                         | They can cause peripheral vasoconstriction, thus limiting cooling, and can affect central thermoregulation.                                                                                           |
| <b>Antihistamines</b>                                                                                                                                                                                                     | They can cause peripheral vasoconstriction, limiting cooling.                                                                                                                                         |
| <b>Antipsychotics</b>                                                                                                                                                                                                     | They interfere with hypothalamic thermoregulation.                                                                                                                                                    |
| <b>Benzodiazepines</b>                                                                                                                                                                                                    | They can reduce alertness and affect judgment and perception of heat.                                                                                                                                 |
| <b>Beta-blockers</b>                                                                                                                                                                                                      | They decrease heart rate and contractility.                                                                                                                                                           |
| <b>Blockers calcium channels</b>                                                                                                                                                                                          | They decrease cardiac contractility and compromise vascular compensatory mechanisms.                                                                                                                  |
| <b>Diuretics</b>                                                                                                                                                                                                          | They can increase the risk of dehydration and hypovolemia.                                                                                                                                            |
| <b>Illegal drugs (e.g.: cocaine, heroin, phencyclidine, MDMA)</b>                                                                                                                                                         | They can increase the production of metabolic heat and reduce alertness and judgment.                                                                                                                 |
| <b>Thyroid hormones</b>                                                                                                                                                                                                   | can increase heat production.                                                                                                                                                                         |
| <b>Laxatives</b>                                                                                                                                                                                                          | They can increase the risk of dehydration and hypovolemia.                                                                                                                                            |
| <b>Lithium</b>                                                                                                                                                                                                            | It can reduce alertness and affect judgment and heat perception, and cause nephrogenic diabetes insipidus; levels can rise to dangerous levels and cause kidney damage in the context of dehydration. |
| <b>Serotonin reuptake inhibitors</b>                                                                                                                                                                                      | They can interfere with hypothalamic thermoregulation.                                                                                                                                                |
| <b>Supplements for weight loss that can increase the metabolic rate (e.g.: carnitine, green tea extract and other types of teas, such as ginger, black, and yerba mate, as well as foods with thermogenic properties)</b> | They can increase heat production.                                                                                                                                                                    |

Source: SMS-Rio. Technical note on the performance of primary care teams in the face of heat waves. Published on 12/15/2023.

# Clinical conditions related to exposure to extreme heat and management

The clinical conditions that can be triggered or aggravated by heat are divided into: conditions directly resulting from thermal stress and dehydration, dermatological conditions related to heat, the onset or exacerbation of chronic conditions, and psychoneurological conditions. Table 12 summarizes the main clinical conditions, guiding their management in primary health care and the corresponding ICD codes.

Heat stress conditions encompass a spectrum of varying severity, ranging from mild cases such as cramps and heat edema to severe cases of heat stroke, in which the patient presents with a body temperature above 40°C and altered level of consciousness, and whose mortality rate, if not adequately treated, can reach 80%. A noteworthy intermediate-risk condition is heat exhaustion, which can initially be difficult to differentiate from an early stage of heat stroke due to the intensity of the discomfort and the possible presence of milder alterations in mental status, such as irritability and impaired judgment.

An important aspect of clinical assessment is the technique for measuring body temperature. Axillary temperature measurement, which is the most frequently used technique, reflects peripheral temperature, which can be very different from core temperature; however, heat-related damage is precisely due to the elevation of core temperature. Thus, if heat stroke is suspected, it is important to use an appropriate technique, such as rectal temperature measurement. If this is not used, a high degree of suspicion of heat stroke should be adopted in the presence of altered levels of consciousness, regardless of the measured body temperature.

Both mild heat stroke and heat exhaustion can, if heat exposure is not removed and hydration is not provided, progress to increasingly severe forms. It is noteworthy that heat stroke occurs predominantly in the vulnerable groups listed in Table 10 and should be monitored even at relatively low heat levels, such as in NC2 and NC3.

Table 12 — Group of general conditions and behaviors

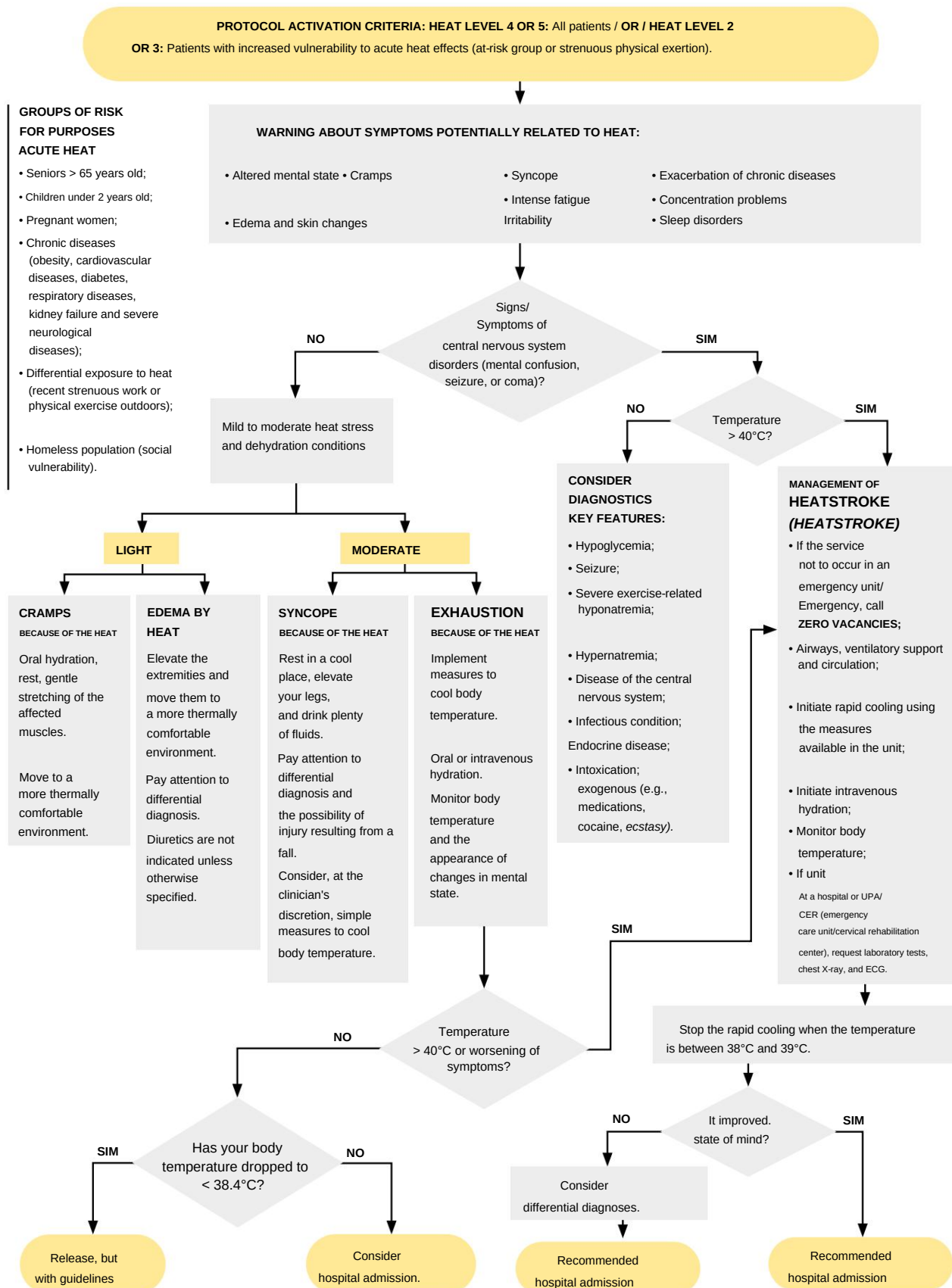
| GROUP OF CONDITIONS                    | CONDITION — CID CODE                                                 | CLINICAL PICTURE                                                                                                                                                                                                                                                                                                              | CONDUCT                                                                                                                                                                        |
|----------------------------------------|----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Heat stress and dehydration conditions | <b>Cramps</b><br><small>because of the heat</small><br><b>(T627)</b> | Painful muscle spasms in areas of high activity, such as the arms, legs, or abdomen.<br>It can also present with cold, wet skin, but normal body temperature. The condition is secondary to neuromuscular hyperactivity and depletion of fluids and electrolytes, which is exacerbated by extreme heat.                       | Oral hydration, rest, gentle stretching of affected muscles. Move to a more thermally comfortable environment.                                                                 |
|                                        | <b>Edema due to heat</b><br><b>(T677)</b>                            | Mild edema resulting from cutaneous vasodilation, leading to an increase in interstitial fluid.<br>It usually affects the ankles and feet after prolonged exposure to heat, without signs of heart failure or a history of other chronic diseases that would justify the edema.<br>Occasionally with facial <i>flushing</i> . | Elevate the extremities and move to a more thermally comfortable environment. Pay attention to differential diagnosis. Diuretics are not indicated unless otherwise specified. |

| GROUP OF CONDITIONS                           | CONDITION — CID CODE                                        | CLINICAL PICTURE                                                                                                                                                                                                                                                                                                                                          | CONDUCT                                                                                                                                                                                                                                                                                   |
|-----------------------------------------------|-------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Conditions stress thermal and dehydration     | <b>Exhaustion</b><br>because of the heat<br><b>(T673)</b>   | Intense fatigue, thirst, dizziness, nausea, and profuse sweating after prolonged exposure to heat, with preserved ability to sweat. The temperature is below 40°C, and there are no significant alterations in consciousness, although there may be mild changes in mental state, such as irritability.<br>and commitment of judgment capacity.           | Oral or intravenous hydration, rest in a cool environment. Consider other cooling measures. Careful monitoring of body temperature and neurological status is required due to the increased risk of heatstroke.                                                                           |
|                                               | <b>Heatstroke</b><br><b>(T670)</b>                          | Body temperature > 40°C, altered consciousness (confusion, agitation, delirium, coma), hot, dry or sweaty skin, which may also progress to seizures and shock. Symptoms may mimic other conditions: sepsis, ischemic stroke, and toxicological emergencies. In a late stage, it can progress to disseminated intravascular coagulation and organ failure. | Management in an emergency unit, preferably a hospital setting. While awaiting transfer to the emergency unit, instruct patients to apply cold compresses to the neck, armpits, and groin area, providing life support as needed.                                                         |
| Related dermatological conditions in the heat | <b>Heat rash, especially miliaria.</b><br><br><b>(L740)</b> | Vesicles or small pustules, accompanied by itching, common in skin folds and under tight clothing, especially those made of synthetic material. The use of creams<br>Oily products (e.g., tanning lotion) can promote the appearance or worsen the condition.                                                                                             | Opt for lightweight, loose-fitting clothing made from natural fibers, such as cotton. Frequent showers are recommended to cool the body. Use neutral soap in the shower.<br>Keep skin cool and folds dry.<br>Consider application starch mixture made from corn and water or water paste. |

| GROUP OF CONDITIONS                           | CONDITION — CID CODE                      | CLINICAL PICTURE                                                                                                                                                                                | CONDUCT                                                                                                                                                                                     |
|-----------------------------------------------|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Related dermatological conditions in the heat | Sunburn (L55)                             | Redness of the skin, sensation<br>Symptoms include heat and pain, with possible blisters in severe cases, after prolonged sun exposure.                                                         | Oral hydration, application of soothing lotions, and avoiding further sun exposure are recommended.                                                                                         |
| Onset or exacerbation of chronic diseases     | Cardiovascular problems                   | Symptoms of angina or evidence of myocardial infarction, such as chest pain, shortness of breath, and sweating in patients with a prior history of cardiovascular disease or with risk factors. | Assessment and management according to the chest pain protocol, in an urgent care and emergency unit. If initially seen in a primary care unit, consider performing an ECG before referral. |
|                                               | Respiratory problems                      | Increased dyspnea, wheezing, and cough in patients with chronic respiratory diseases, especially on days of heat and pollution.                                                                 | Administration of bronchodilators, oxygen therapy, humidification of the airways, assessment for hospitalization, if necessary.                                                             |
|                                               | Kidney problems                           | Dehydration can compromise kidney physiology, worsening the condition of base or inducing a acute kidney injury.                                                                                | Hydration and monitoring of the painting.                                                                                                                                                   |
| Conditions psychoneurological                 | Concentration problems                    | Difficulty maintaining focus, slow thinking, mental fatigue during exposure to heat.                                                                                                            | Advice to avoid excessive heat, adequate hydration, and guidance on adjustments in the work/study environment.                                                                              |
|                                               | Disorders where are they and irritability | Insomnia or interrupted sleep, increased irritability during periods of intense heat.                                                                                                           | Measures to cool the sleeping environment, sleep hygiene, assessment for stress or anxiety.                                                                                                 |

Figure 4 presents a flowchart for the management protocol of heat-related clinical changes, focusing on early detection and management of heat stroke.

**Figure 4 — Flowchart for managing heat-related changes in the municipal health network of Rio de Janeiro**



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## APPENDICES

### ANNEX 1. RIO DECREE No. 54,740, OF JUNE 28, 2024



Protocol: 996199  
Data: 01/07/2024  
Title: RIO DECREE No. 54740  
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#### RIO DECREE No. 54740 OF JUNE 28, 2024

This law establishes the classification of Heat Levels in risk scenarios related to extreme heat, for the purpose of defining action protocols within the municipality, and provides other measures.

**The Mayor of the City of Rio de Janeiro**, in the exercise of the powers conferred upon him by current legislation, and

Considering the need to develop and implement an action protocol to combat the city's climate risks, in accordance with the Sustainable Development and Climate Action Plan of the City of Rio de Janeiro - PDS, established by Decree Rio No. 48,940, of June 4, 2021;

Considering the need to improve the situational awareness of the population of Rio de Janeiro by simplifying the communication of the city's heat levels, and thus promote a culture of safety in the face of risk scenarios related to extreme heat;

Considering the need to ensure the effective implementation of city operation management in risk scenarios related to extreme heat, seeking efficient responses from various government agencies at different levels of government, public service concessionaires, and companies that provide services to the population of the city of Rio de Janeiro;

Considering the need to promote resilience and increase the degree of adaptation to risk, urgency and emergency scenarios with integrated rapid response actions to the climate risk of extreme heat;

Considering Decree Rio No. 53,105, of August 23, 2023, which *provides for the authorization of events in public and private areas in the municipality of Rio de Janeiro*,

#### DECREE:

**Article 1.** The Heat Level (HL) is hereby established, with the purpose of defining guidelines for action for various governmental bodies at different levels of government, public service concessionaires, and companies that provide services within the territory of the city of Rio de Janeiro, and for communication with citizens, as a trigger parameter for addressing the climate risk of extreme heat.

**Article 2.** For the purposes of this Decree, the following definitions apply:

I - Heat Index - HI: a measure to define the intensity of heat varying as a function of temperature and relative humidity of the air;

II - Risk scenarios related to extreme heat: circumstances with the possibility of aggravation, considering a scenario with the potential to cause injury, illness or death of people, damage to property, the environment and image, or a combination of these factors, related to the effects of the Extreme Heat climate risk defined in the PDS;

III - Heat Index Forecast: Numerical forecast models will be adopted for temperature, humidity, and heat index estimates for three days, updated every 4 (four) hours.

IV - Activity at risk for extreme heat: work activities carried out in outdoor areas subject to solar radiation, without air conditioning, refrigeration or adequate shading, or in areas with a concentration of people.

V - Heat Waves: 3 (three) consecutive days with an IC greater than 36°C (thirty-six degrees Celsius) up to 44°C (forty-four degrees Celsius), for at least 4 (four) hours per day or greater than 44°C (forty-four degrees Celsius), for at least 2 (two) hours;

VI - Triggering Parameters: indicators established by experts to measure the level of specific risks, by thematic areas, with the potential to trigger significant changes in the city's daily routine and that guide decision-making for changes in operational stage.

**Article 3.** For the purpose of establishing the NC, the relationship between the IC and the time of exposure to heat will be considered.

**Article 4.** The NC will be classified into 5 (five) risk levels, as follows:

Level I - Heat 1: No high heat levels are expected. At this level, the city continues with its normal routine.

II - Heat Level 2: Forecast or record of high heat levels for one or two consecutive days.

III - Heat Level 3: Recording of high heat indices with a forecast of persistence or increase for at least three consecutive days.

IV - Heat Level 4: Recording of very high heat indices with a forecast of persistence or increase for at least three consecutive days. At this level, adaptation of activities at risk for extreme heat may be indicated, seeking to protect the population from the impacts of heat waves.

V - Heat Level 5: Recording of extreme heat indices with a forecast of persistence or increase for at least three consecutive days. At this level, activities at risk from extreme heat that are not adapted to cope with the heat wave may be recommended for suspension in order to protect the population from the impacts of heat waves.

**Article 5.** The Operations and Resilience Center - GP/COR will be responsible for monitoring and disseminating the Heat Levels.

**Article 6.** The oversight of compliance with the adaptation measures and the interruption of activities at risk due to extreme heat and public gatherings, as stipulated in this Decree, shall be the responsibility of:

I - Executive Advisory Services for Events and Major Events in the City of Rio de Janeiro - GP/AEGE;

II - Municipal Secretariat of Public Order - SEOP.

**Article 7.** The Committee for the Development of Protocols for Coping with Extreme Heat - CDPECE is hereby created, which will be responsible for defining coping protocols for risk scenarios related to extreme heat.

**Article 8.** The CDPECE will be composed of representatives appointed by the respective heads of the following bodies:

I - Operations and Resilience Center - GP/COR;

II - Municipal Health Department - SMS;

III - Municipal Secretariat for the Environment and Climate - SMAC;

§ 1 The CDPECE may request the nomination of public agents from other municipal bodies or entities that have a thematic link with the subject under discussion by the committee, without prejudice to requesting the nomination of employees from other entities or inviting members and entities of civil society that may contribute to the development of the work.

§ 2 The GP/COR will publish an Ordinance designating the members of the CDPECE.

§ 3 The CDPECE will be chaired by the representative appointed by the head of the GP/COR, who will be supported by the technical coordinator appointed by the head of the SMS.

§ 4 The members of the CDPECE will not receive any kind of remuneration for their work, as the performance of their duties is considered to be of significant public interest.

§ 5. Representatives from other bodies and entities may be included in the CDPECE, to be invited by the GP/COR.

§ 6 Other bodies may request meetings with CDPECE for the development or updating of protocols that are relevant to their areas of activity.

**Article 9.** All actions of the CDPECE will be carried out through an Order from the GP/COR.

**Article 10.** The CDPECE will publish, through a GP/COR Ordinance, its own act regulating this Decree.

**Article 11.** Municipal bodies shall provide for the revision of their operational and administrative action protocols, based on the classification of Heat Levels provided for in Article 4, and forward them to the GP/COR within a maximum period of thirty days, counted from the publication of the normative act described in Article 10.

**Article 12.** This Decree shall enter into force on the date of its publication.

Rio de Janeiro, June 28, 2024; 460th anniversary of the founding of the City.

**EDUARDO PAES**

# ANNEX 2. OPERATIONS AND RESILIENCE CENTER — ORDINANCE “N” No. 03/2024, OF JUNE 28, 2024



Protocol: 996184  
Data: 01/07/2024  
Title: Regulation on Heat Levels v3  
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## OPERATIONS AND RESILIENCE CENTER PORTARIA "N" N°03/2024 DE 28 DE JUNHO DE 2024

**Regulates Heat Levels in scenarios of risk related to extreme heat for the definition of action protocols within the Municipality, and provides other measures.**

**The Chief Executive of the Operations and Resilience Center of the Mayor's Office**, in the exercise of the powers conferred upon him by current legislation,

Considering the provisions of Rio Decree No. 33,322, of December 23, 2010, which creates the Integrated Operational Control Center of the Rio de Janeiro City Hall through the Operations and Resilience Center;

Considering the provisions of Rio Decree No. 54,740 of June 28, 2024, which provides for the classification of Heat Levels within the Municipality, and takes other measures;

### **RESOLVE:**

**Article 1.** The Operations and Resilience Center of the Mayor's Office - GP/COR will define, through an indication of the Heat Level defined by a tool developed in partnership with the Municipal Health Department - SMS, the need to change the risk scenario related to extreme heat in the city, seeking to strengthen the public administration's capacity to react to situations that impact the city's routine and/or have the potential to worsen;

**Article 2.** For the purposes of this decree, the following definitions apply:

I - City Operational Stages: risk classification strategy used by GP/COR to qualify the city's scenario, considering the composition of various factors present and providing a general overview of anomalies, taking into account their aggravation parameters and the perspective of the future scenario.

II - Risk Scenarios related to extreme heat: circumstances with the potential for aggravation, considering a scenario with the potential to cause injury, illness or death of people, damage to property, the environment and image, or a combination of these factors, related to the effects of the Extreme Heat climate risk defined in the PDS;

III - Bodies: the parts that make up the Executive Branch resulting from the decentralization of its administrative function, including secretariats, coordinating bodies, public companies, foundations, and autonomous agencies that operate in the city of Rio de Janeiro;

IV - Agencies: a generic term encompassing governmental bodies from different levels of government, public service concessionaires, and companies that provide services to the population of the city of Rio de Janeiro;

V - Times of greatest heat exposure: Times when historically the highest IC values are recorded during the day;

VI - Event: For the purposes of this Decree, an event is considered to be any temporary activity of an economic, cultural, sporting, recreational, musical, artistic, exhibition, civic, commemorative, social, religious or political nature, whether for profit or not, that generates a concentration or influx of people, in open or closed areas, whether private or not;

VII - Medium-sized events: between 2,001 (two thousand and one) and 10,000 (ten thousand) people;

VIII - Large-scale events: between 10,001 (ten thousand and one) and 50,000 (fifty thousand) people;

IX - Mega-events: above 50,000 (fifty thousand) people.

**Article 3.** Actions to address risk scenarios related to extreme heat will be coordinated by the relevant agencies and managed by the GP/COR, which is responsible for monitoring the tool, with the aim of reducing the impacts caused by these scenarios.

**Article 4.** The GP/COR shall monitor and, if necessary, report changes in risk scenarios related to extreme heat to the agencies, so that they may initiate the action protocols for the agreed Heat Levels.

**Article 5.** The following measures shall be adopted in accordance with the NC, as described below:

I - HEAT 1 - level at which an IC is recorded that is less than or equal to 36°C (thirty-six degrees Celsius) or when it is greater than 36°C (thirty-six degrees Celsius), for a period of less than 4 (four) hours per day;

- a) The COR will maintain monitoring of Heat Levels.
- b) The COR may coordinate and disseminate information on heat forecasts for the coming days.

II - HEAT 2 - level in which there is a record of up to 2 (two) consecutive days, with an IC above 36°C (thirty-six degrees Celsius), for at least 4 (four) hours per day;

- a) The COR will maintain monitoring of Heat Levels;
- b) The COR (Operations Center) should coordinate and disseminate information about the symptoms of heat exposure according to the levels, and guide the population with the technical support of the SMS (Municipal Health Secretariat).

III - HEAT 3 - level in which there is a recorded IC between 36°C (thirty-six degrees Celsius) and up to 40°C (forty degrees Celsius), for at least 4 (four) hours per day. In addition, a forecast or record of 2 (two) more days with IC greater than 36°C, for 4 (four) hours per day. The period must total at least 3 (three) consecutive days;

- a) The COR will maintain monitoring of Heat Levels;
- b) Coordinated by the COR and with the technical support of the SMS, other agencies should reinforce the dissemination of information about the symptoms of heat exposure and guide the population.

IV - HEAT 4 - level in which there is a recorded IC between 40°C (forty degrees Celsius) and up to 44°C (forty-four degrees Celsius), for at least 4 (four) hours per day. In addition, a forecast or record of 2 (two) more days with IC greater than 36°C, for 4 (four) hours per day. The period must total at least 3 (three) consecutive days;

- a) The COR will maintain monitoring of Heat Levels;
- b) Coordinated by the COR and with the technical support of the SMS, other agencies should reinforce the dissemination of information about the symptoms of heat exposure and guide the population.
- c) Air-conditioned or refrigerated locations will be designated to serve as cooling points for the population and will be defined by the Committee for the Development of Protocols for Coping with Extreme Heat - CDPECE, within 60 days;
- d) The COR will inform the public about the open cooling points;
- (e) The operating hours of public places with air conditioning or refrigeration or shaded areas may be extended, as defined by CDPECE, within 60 days;
- f) The periods for body cooling and hydration may be extended for work activities performed in outdoor areas subject to solar radiation, without air conditioning, refrigeration or adequate shading, considering 15 (fifteen) minutes of break for every 45 (forty-five) minutes of activity;

- (g) The availability of hydration or water distribution stations will be expanded in the areas where the most vulnerable populations are housed;
- h) Medium-sized, large-scale, or mega-events in outdoor areas exposed to solar radiation and with large gatherings of people may be cancelled or rescheduled if, at the very least, the programming is not adjusted to avoid attractions during peak heat exposure times.

and they don't even offer free access to water.

- i) Activities carried out in outdoor areas exposed to solar radiation in schools will be suspended and transferred to shaded or indoor areas;
- j) It is recommended that companies and entities adopt teleworking or remote work arrangements for their employees, reducing exposure to solar radiation during their commute;

V - HEAT 5 - level in which there is a record of IC above 44°C (forty-four degrees Celsius), for at least 2 (two) hours per day. In addition, forecast or records of 2 (two) more days with IC greater than 36°C, for 4 (four) hours per day. The period must total at least 3 (three) consecutive days;

- a) The GP/COR will maintain monitoring of Heat Levels;
- b) Coordinated by GP/COR and with technical support from SMS, other agencies should reinforce the dissemination of information about the symptoms of heat exposure and guide the population. CDPECE will release meteorological bulletins every 6 hours, in addition to an epidemiological bulletin within 72 hours after the end of the heat wave.

c) Locations with air conditioning or refrigeration will be designated to serve as cooling points for the population and will be defined by CDPECE within 60 days;

d) The GP/COR will inform the public about the open cooling points;

(e) The operating hours of public places with air conditioning or refrigeration or shaded areas may be extended, as defined by CDPECE within 60 days;

f) The periods for body cooling and hydration may be extended for work activities performed in outdoor areas subject to solar radiation, without air conditioning, refrigeration or adequate shading, considering 15 (fifteen) minutes of break for every 45 (forty-five) minutes of activity;

(g) The availability of hydration or water distribution stations will be expanded in the areas where the most vulnerable populations are housed;

h) Medium-sized, large-sized, or mega-events in outdoor areas exposed to solar radiation and with a concentration of people may be canceled or rescheduled if they do not, at a minimum, adjust their programming to avoid attractions during peak heat exposure times, offer free access to water, reduce queue times, provide shaded areas, or offer covered cooling areas with sprinklers or air conditioning.

i) Outdoor activities at schools will be suspended and moved indoors;

j) It is recommended that companies and entities adopt teleworking or remote work arrangements for their employees, reducing exposure to solar radiation during their commute;

k) Tourist attractions lacking adequate air conditioning, refrigeration, or shade will be closed;

l) Suspend activities carried out in outdoor areas subject to solar radiation during peak heat exposure hours, except in areas critical to the city's operation.

**Article 6.** The GP/COR shall use the Heat Levels as one of the Deflagration Parameters for changing the Operational Stages defined in Rio Decree No. 53525, of November 15, 2023.

**Article 7.** Municipal bodies shall be responsible for maintaining the updating of their operational and administrative action protocols, as provided for in Article 11 of Rio Decree No. 54,740 of June 28, 2024.

**Article 8.** This decree enters into force on the date of its publication.



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